Types of Variables in Python

Type	Draft
Status	Not started
Project	My Python Journey

In Python, variables can be classified in different ways. For beginners, first understand these 4 important types based on scope and usage.

1. Local Variable

A variable created inside a function is called a local variable.

It can normally be accessed only inside that function.

```
def student():    name = "Gaythri"    print(name)student()
```

Output:

```
Gaythri
```

Here, `name` is a local variable because it is created inside the function.

2. Global Variable

A variable created outside all functions is called a global variable.

It can be accessed inside and outside functions.

```
name = "Gaythri"def student():    print(name)student()print(name)
```

Output:

```
Gaythri
Gaythri
```

Here, `name` is a global variable.

3. Instance Variable

An instance variable belongs to a particular object of a class.

```
class Student:    def __init__(self, name):        self.name = names1 = Student("Gaythr
i")s2 = Student("Priya")print(s1.name)print(s2.name)
```

Output:

```
Gaythri
Priya
```

Here, `name` is an instance variable. Each object can have its own value.

4. Class Variable

A class variable is declared inside a class but outside its methods. It is shared by the objects of that class unless an object overrides it.

```
class Student:    college = "SVERI"s1 = Student()s2 = Student()print(s1.college)print(s2.
college)
```

Output:

```
SVERI
SVERI
```

Here, `college` is a class variable.

Difference between all four

Type	Declared where?	Belongs to
Local	Inside a function	Function
Global	Outside functions	Module/program
Instance	Inside a class method, usually using `self`	Individual object
Class	Inside class, outside methods	Class

Important interview point

Local and global variables are related to scope.

Instance and class variables are related to object-oriented programming (OOP).

Practice question

What will be the output of this program?

```
x = 100def display():    x = 50    print(x)display()print(x)
```

Choose your answer:

100, 100

50, 100

50, 50

Error

Check answer